START: Daily Scheduler
Triggers at 04:00 AM
4 Cities Queue

STEP 1: Data Retrieval
Query GEE API: MODIS, S2, S5P + ERA5
Apply Look-back: scan for newly available past dates
→ Append to D1

ERA5 data available up to today?

NO

ERA5-LSTM Sub-Model
Feed last 45-day ERA5
LSTM predicts ERA5 → today
Fills 7-8 day lag

YES

STEP 2: Tri-Sat Prediction
For MODIS, S2, S5P:
If raw in D1 → use as-is | If missing → predict
Tree-based models per satellite per city

STEP 3: Build Temp Dataset
Merge: raw D1 + ERA5-pred + Tri-Sat pred
Rolling 30-day window per city → D2

STEP 4: Feature Engineering
Compute: AOD, NDVI, UV index, BLH, RH
Generate: lag features, rolling stats
Temporal encoding → Feature Matrix

STEP 5: Ensemble Prediction
Run: RF · XGBoost · LightGBM · CatBoost
Stack meta-learner → best per city
→ Today's PM2.5 to D3

STEP 6: LSTM Forecast
Load 30-45d from D3 + D2
Re-apply Feature Engineering
LSTM → 7-day forecast per city

STEP 7: Frontend Update
Today's PM2.5 · 30-day history
7-day forecast · City navigation
Render per-city page

All 4 cities processed?

NO
next city

YES

PIPELINE COMPLETE
Idle until next 04:00 AM trigger